PROJECT: G+3P+19 FLOORS+ ROOF - COMMERCIAL & RESIDENTIAL BUILDING,
PLOT NO. 3460608 @ BUSINESS BAY, DUBAI

CLIENT	CONSULTANT	CONTRACTOR	MEP SUBCONTRACTOR
KASCO REAL ESTATE DEVELOPMENT L.L.C	NEXT ENGINEERING CONSULTANTS	LA MIDA CONTRACTING L.L.C	ELMA ELECTROMECHANICAL CO.LLC

SD REF. NO.	REV.:	SHOP DRAWING	DATE
ONDA-AC-SD-MEP-VEN-006	0	SUBMITTAL	12/11/2025

Submittal Stamp (Contractor)

To Engineer: NEXT ENGINEERING CONSULTANTS

Submittals Status: 1st submittal Re-submittal (Specify Rev-02.)

Subject: VENTILATION DRAWING

B.O.Q Item/Specs.: _

Others (Specify):

Drawing Nos.	Rev.	Date	Copies	Description
LMC-ONDA-AC-SD-VEN-306	0	12/11/2025	1	2ND TO 18TH FLOOR VENTILATION LAYOUT

Submittal Stamp (Sub-Contractor)

Received Stamp (Consultant)

Shop Drawings are submitted for your review / approval:

MEP Contractor's Stamp & Sign	Contractor's Stamp & Sign
Eng. Sajan John Signature: Date: 12/11/2025	Eng.Moustafa Elshorbagy - Sr. Projects Manager Signature: Date:

Engineer's Comment:

Reply Stamp (Consultant)

Checked by: (Signature)

Projects Manager: **Eng. Yahia Shamsia** (Signature) Date

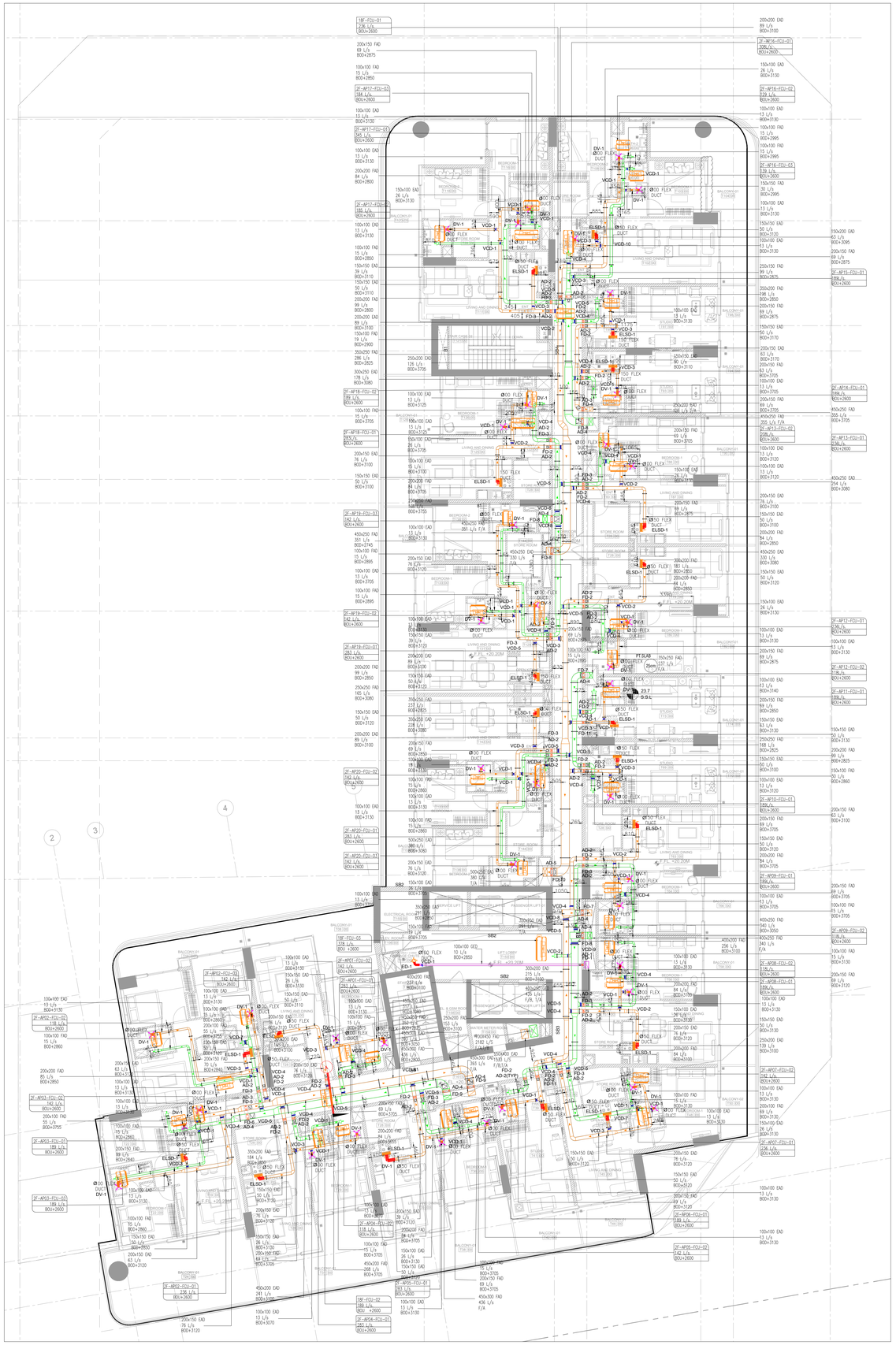
Received Stamp (Contractor)

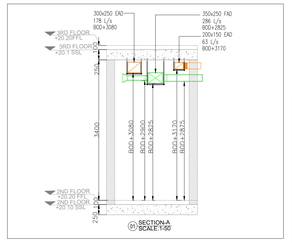
The afore mentioned shop drawing has been:

☐ A	☐ B	☐ C	☐ D
Approved	Approved,Subject to Comments	Revise & Resubmit	Rejected

Received by Contractor: (Please sign below. Retain Original & Return Copy)
Name: Date:

* The CONTRACTOR shall not proceed with the relevant WORKS if any cost or time implication is involved, UNLESS relevant notification pursuant to Clause 51.2 & 52.2 of the conditions of contract is submitted by the Contractor followed by WRITTEN instruction by the ENGINEER to proceed with the subject works.





SECTION-A
SCALE:1:50

DISK VALVE			
REF #	NECK SIZE DIA (mmmmm)	AIR FLOW L/s	QTY.
DV-01	ø100	1.3L/s	41

SCHEDULE OF FIRE DAMPER (FD)			
SL.NO	TAG NO.	QTY.	SIZE (mm)
01	FD-01	01	100X100
02	FD-02	19	200X150
03	FD-03	16	200X200
04	FD-04	01	250X200
05	FD-05	01	300X200
06	FD-06	02	350X200
07	FD-07	02	350X250
08	FD-08	04	450X250
09	FD-09	02	450X300
10	FD-10	01	500X250
11	FD-11	02	150X150

SCHEDULE OF DUCT INSULATION (WITH SELF ADHESIVE)			
SL.NO	DESCRIPTION	INSULATION THICKNESS WITHIN CONDITIONED AREA	INSULATION THICKNESS WITHIN NON CONDITIONED AREA
1	Supply air / Return air Duct	13mm	25mm
2	Treated Outside Fresh Air Duct	13mm	25mm
3	Air Conditioned General Extract Air Duct	13mm	25mm
4	Transfer Untreated Outside Air Duct	25mm	Un-Insulated

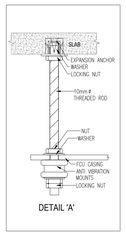
SCHEDULE OF ACCESS DOOR			
S.NO.	TAG NO.	SIZE (MM)	QTY Nos
1	AD-01	100x200	02
2	AD-02	150x200	34
3	AD-03	200x200	02
4	AD-04	300x300	09
5	AD-05	450x450	01

SCHEDULE OF FLEXIBLE DUCTS		
S.NO	FLEXIBLE DUCT SIZE	MAXIMUM AIR FLOW
1	ø100	15 L/s
2	ø150	35 L/s
3	ø200	65 L/s
4	ø250	100 L/s
5	ø300	150 L/s

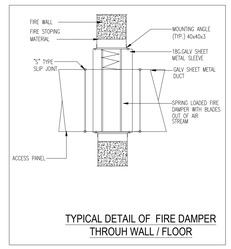
SCHEDULE OF VOLUME CONTROL DAMPER(VCD)		
TAG NO.	QTY.	SIZE (mm)
VCD-1	32	100X100
VCD-2	07	150X100
VCD-3	14	150X150
VCD-4	29	200X150
VCD-5	13	200X200
VCD-6	01	250X250
VCD-7	03	200X100
VCD-8	02	300X200
VCD-9	01	400X200

SCHEDULE OF EXTRACT DIFFUSER							
S.NO.	TAG NO.	TYPE	AIR FLOW (L/S)	NOISE (DB)	VELOCITY (M/S)	SIZE (MM)	QTY. (NOS.)
1	ED-01	EXTRACT AIR DIFFUSER	10	23	2	150X150	01

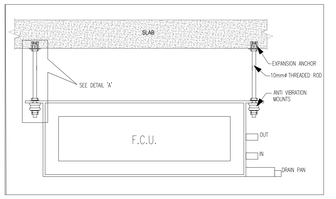
SCHEDULE FOR LINEAR SLOT DIFFUSER								
S.NO	TAG NO.	TYPE	SIZE (MM)	LENGTH (MM)	AIR FLOW (L/S)	QTY. (NOS.)	NOISE (DB)	VELOCITY (M/S)
1	ELSD-01	EXTRACT LINEAR SLOT DIFFUSER	25SLOTS 25MM	500	50	20	23	2.0

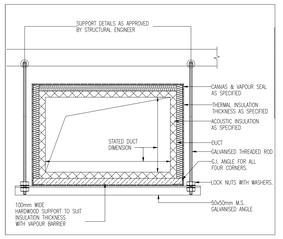


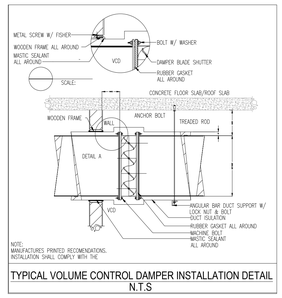
DETAIL 'A'



TYPICAL DETAIL OF FIRE DAMPER THROUH WALL / FLOOR







TYPICAL VOLUME CONTROL DAMPER INSTALLATION DETAIL
N.T.S

FCU SCHEDULE

UNITS REF.	MODEL	TOTAL LOAD KW	SENSIBLE LOAD KW	AIR FLOW LPS	ESP (PA)	AIR ON COIL DB/WB°C	AIR OFF COIL DB/WB°C	WATER FLOWRATE LPS	WKPA	FANVOLTAGE V	UNIT TYPE	POWER I/P WATTS	DIMENSIONS (MM) H	L	W	Weight kg	QTY
2F-AP01-FCU-01	FWW800VAN	5.22	4.14	315	50	23.9/17.1	11.8/11.4	0.14	13	7.3	Ducted	119	235	1425	536	33.4	1
2F-AP01-FCU-02	FWW400VAN	2.64	2.22	160	50	23.7/16.9	11.6/11.4	0.07	15	8.5	Ducted	50	235	865	536	19	1
2F-AP02-FCU-01	FWW700VAN	4.28	3.38	271	50	23.6/16.8	11.7/11	0.11	20	7.2	Ducted	109	235	1095	536	23.2	1
2F-AP02-FCU-02	FWW300VAN	2.23	1.76	129	50	23.9/17.1	11.7/11.3	0.06	11	8.5	Ducted	40	235	815	536	17.4	1
2F-AP02-FCU-03	FWW500VAN	2.99	2.39	184	50	23.8/16.8	11.7/11	0.08	10	6.6	Ducted	77	235	945	536	20.4	1
2F-AP03-FCU-01	FWW600VAN	4.08	3.17	245	50	23.8/17.2	11.9/11.5	0.11	19	8.5	Ducted	87	235	1045	536	21.9	1
2F-AP03-FCU-02	FWW400VAN	2.57	2.22	160	50	23.6/16.7	11.5/11.3	0.07	15	8.5	Ducted	50	235	865	536	19	1
2F-AP03-FCU-03	FWW500VAN	3.18	2.76	216	50	23.6/16.5	11.7/11.2	0.08	11	8.5	Ducted	77	235	945	536	20.4	1
2F-AP04-FCU-01	FWW700VAN	4.63	3.78	296	50	23.9/17.1	11.9/11.6	0.12	23	8.5	Ducted	109	235	1095	536	23.2	1
2F-AP04-FCU-02	FWW300VAN	2.21	1.76	129	50	23.8/17.2	11.7/11.3	0.06	10	8.5	Ducted	40	235	815	536	17.4	1
2F-AP05-FCU-01	FWW700VAN	4.45	3.72	296	50	23.6/16.8	11.8/11.4	0.12	22	8.5	Ducted	109	235	1095	536	23.2	1
2F-AP05-FCU-02	FWW400VAN	2.64	2.22	160	50	23.7/16.9	11.6/11.4	0.07	15	8.4	Ducted	50	235	865	536	19	1
2F-AP06-FCU-01	FWW600VAN	3.9	3.17	245	50	23.6/16.8	11.7/11.2	0.1	17	8.5	Ducted	87	235	1045	536	21.9	1
2F-AP07-FCU-01	FWW700VAN	4.55	3.69	296	50	23.6/17	11.9/11.6	0.12	23	8.5	Ducted	109	235	1095	536	23.2	1
2F-AP07-FCU-02	FWW500VAN	3.07	2.37	183	50	23.9/17.1	11.9/11.2	0.08	11	6.5	Ducted	77	235	945	536	20.4	1
2F-AP08-FCU-01	FWW600VAN	3.89	3.14	245	50	23.6/16.9	11.8/11.3	0.1	17	8.4	Ducted	87	235	1045	536	21.9	1
2F-AP08-FCU-02	FWW300VAN	2.21	1.75	129	50	23.7/17	11.7/11.3	0.06	10	8.5	Ducted	40	235	815	536	17.4	1
2F-AP09-FCU-01	FWW600VAN	4	3.15	245	50	23.7/17.1	11.9/11.5	0.11	18	8.4	Ducted	87	235	1045	536	21.9	1
2F-AP09-FCU-02	FWW300VAN	2.21	1.75	129	50	23.7/17	11.7/11.3	0.06	10	8.5	Ducted	40	235	815	536	17.4	1
2F-AP10-FCU-01	FWW500VAN	3.39	2.76	216	50	23.9/17.1	12/11.6	0.09	13	8.4	Ducted	77	235	945	536	20.4	1
2F-AP11-FCU-01	FWW600VAN	3.95	3.18	245	50	23.7/16.9	11.7/11.2	0.11	18	8.5	Ducted	87	235	1045	536	21.9	1
2F-AP12-FCU-01	FWW700VAN	4.63	3.78	296	50	23.9/17.1	11.9/11.6	0.12	23	8.4	Ducted	109	235	1095	536	23.2	1
2F-AP12-FCU-02	FWW400VAN	2.4	1.89	141	50	23.6/16.8	11.5/11	0.06	13	7.2	Ducted	50	235	865	536	19	1
2F-AP13-FCU-01	FWW700VAN	4.63	3.78	296	50	23.9/17.1	11.9/11.6	0.12	23	8.4	Ducted	109	235	1095	536	23.2	1
2F-AP13-FCU-02	FWW400VAN	2.75	2.23	160	50	23.8/17.2	11.7/11.6	0.07	17	8.4	Ducted	50	235	865	536	19	1
2F-AP14-FCU-01	FWW600VAN	3.95	3.21	245	50	23.8/16.9	11.7/11.2	0.11	18	8.5	Ducted	87	235	1045	536	21.9	1
2F-AP15-FCU-01	FWW500VAN	3.39	2.76	216	50	23.9/17.1	12/11.6	0.09	13	8.5	Ducted	77	235	945	536	20.4	1
2F-AP16-FCU-01	FWW400VAN	5.05	4	308	50	23.7/17	11.7/11.2	0.13	12	7.1	Ducted	119	235	1425	536	33.4	1
2F-AP16-FCU-02	FWW300VAN	2.23	1.76	129	50	23.9/17.1	11.7/11.3	0.06	11	8.5	Ducted	40	235	815	536	17.4	1
2F-AP16-FCU-03	FWW400VAN	2.43	1.87	139	50	23.7/17	11.6/11.1	0.06	13	7	Ducted	50	235	865	536	19	1
2F-AP17-FCU-01	FWW800VAN	5.34	4.49	345	50	23.7/17	11.8/11.6	0.14	13	8.4	Ducted	119	235	1425	536	33.4	1
2F-AP17-FCU-02	FWW500VAN	3.03	2.38	185	50	23.7/16.9	11.8/11.1	0.08	10	6.6	Ducted	77	235	945	536	20.4	1
2F-AP17-FCU-03	FWW500VAN	3.02	2.37	184	50	23.7/16.9	11.8/11.1	0.08	10	6.6	Ducted	77	235	945	536	20.4	1
2F-AP18-FCU-01	FWW800VAN	5.27	4.51	345	50	23.6/16.8	11.7/11.5	0.14	13	8.4	Ducted	119	235	1425	536	33.4	1
2F-AP18-FCU-02	FWW500VAN	3.24	2.53	198	50	23.9/17.1	12/11.4	0.09	12	7.3	Ducted	77	235	945	536	20.4	1
2F-AP19-FCU-01	FWW800VAN	5.44	4.52	345	50	23.8/17.1	11.8/11.6	0.14	14	8.4	Ducted	119	235	1425	536	33.4	1
2F-AP19-FCU-02	FWW500VAN	2.9	2.36	183	50	23.6/16.6	11.6/10.9	0.08	10	6.5	Ducted	77	235	945	536	20.4	1
2F-AP19-FCU-03	FWW400VAN	2.7	2.26	160	50	23.9/17	11.6/11.5	0.07	16	8.5	Ducted	50	235	865	536	19	1
2F-AP20-FCU-01	FWW800VAN	5.34	4.49	345	50	23.7/17	11.8/11.6	0.14	13	8.4	Ducted	119	235	1425	536	33.4	1
2F-AP20-FCU-02	FWW400VAN	2.65	2.24	160	50	23.8/16.9	11.6/11.4	0.07	16	8.5	Ducted	50	235	865	536	19	1
2F-AP20-FCU-03	FWW400VAN	2.7	2.26	160	50	23.9/17	11.6/11.5	0.07	16	8.5	Ducted	50	235	865	536	19	1
18F-FCU-01	FWW700VAN	4.64	3.66	296	50	23.6/17.2	12/11.7	0.12	23	8.4	Ducted	109	235	1095	536	23.2	1
18F-FCU-02	FWW600VAN	4.12	3.13	245	50	23.7/17.3	11.9/11.5	0.11	19	8.5	Ducted	87	235	1045	536	21.9	1
18F-FCU-03	FWW1000VAN	6.21	5.28	426	50	23.6/16.7	11.8/11.4	0.16	18	8.5	Ducted	159	235	1475	536	35.9	1

2ND TO 18TH FLOOR VENTILATION LAYOUT - OVER ALL

SCALE-1:100

KEY PLAN



GENERAL NOTES/LEGENDS:

1. ALL DIMENSIONS ARE IN MILLIMETERS,UNLESS OTHERWISE INDICATED.
2. ALL DUCT LEVELS ARE CONSIDERED FROM GROUND FLOOR FFL+15.30mm.

LEGENDS	
SYMBOL	DESCRIPTION
	FRESH AIR DUCT (FAD)
	EXHAUST AIR DUCT (EAD)
	GARBAGE EXHAUST DUCT (GED)
	DUCT SUPPORT
	DISC VALVE (DV)
	FAN COIL UNIT (FCU)
FD	FIRE DAMPER (FD)
VCD	VOLUME CONTROL DAMPER
	FLOW DIRECTION
	FLEX DUCT
	ACCESS DOOR
	SUPPLY DIFFUSER
	RETURN DIFFUSER
	EXTRACT LINEAR SLOT DIFFUSER
FAD	FRESH AIR DUCT
FD	FIRE DAMPER
VCD	VOLUME CONTROL DAMPER
L/s	LITER PER SECOND
FFL	FINISHED FLOOR LEVEL
BOU	BOTTOM OF UNIT
BOD	BOTTOM OF DUCT
FCU	FAN COIL UNIT
FC	FALSE CEILING
F/A	FROM ABOVE
T/A	TO ABOVE
F/B	FROM BELOW
T/B	TO BELOW

DRAWING REFERENCE

NO	DRG: NO	TITLE	REV
01	M-VEN-06-008	2ND FLOOR PLAN VENTILATION LAYOUT	00
02	23491-AR-108	23491-AR-108-2ND FLOOR PLAN	00
03	S-307	THIRD FLOOR PLAN SLAB LAYOUT	00

SHOP DRAWING REVISIONS

NO	DATE	TITLE	REV	STATUS
01	01.11.2025	ISSUED FOR APPROVAL	00	

CLIENT :

KASCO DEVELOPMENTS

CONSULTANT :

NEXT ENGINEERING CONSULTANTS
UAE: Dubai Silicon Oasis
نيكست للاستشارات الهندسية

MAIN CONTRACTOR :

LAMIDA CONTRACTING L.L.C

MEP CONTRACTOR :

ELMA Electromechanical Co. (L.L.C)
P.O.BOX- 126945, DUBAI, U.A.E
TEL: 04 331 2 575, FAX: 04 347 6 275
E-Mail: info@elmaengg.com
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MEP CONTRACTING | ELV SYSTEM | INFRASTRUCTURE

PROJECT TITLE:

COMMERICAL AND RESIDENTIAL BUILDING
G+3P+19FLOORS+ROOF

DRAWING TITLE:

2ND TO 18TH FLOOR
VENTILATION LAYOUT
OVER ALL

Area	Plot No.	
BUSINESS BAY	3460608	
Sheet No. 1	Drawn ANAS	Checked ENG. SADDAM / Approved ENG. SADDAM
Date: 01/11/2025	Scale 1:100@A0	Status SHOP DRAWING
Drawing No. LMC-ONDA-AC-SD-VEN-306		Rev 00